

Informed forcing for CONUS-wide NWM

A user guide to the data—not just the files

1 km

NWM target grid

1 hour

Time resolution

2 streams

NRT + Retro

Best available evidence • Physically consistent fields • Traceable revisions

Better informed—not simply higher resolution

Four design goals shape every production choice.

Use observations

Prefer credible measurements; retain model coverage.

Respect terrain

Adjust meteorology to NWM elevation.

Keep consistency

Temperature, pressure, humidity and longwave stay coupled.

Explain changes

Record sources, quality checks and revisions.

A finer grid is not proof of greater accuracy.

Eight fields, one model-ready product

Hourly meteorology on the NWM grid; daily files for efficient access.

RAINRATE

Precipitation

$\text{kg m}^{-2} \text{s}^{-1}$

T2D

2-m temperature

K

Q2D

2-m specific
humidity

kg kg^{-1}

PSFC

Surface pressure

Pa

SWDOWN

Downward
shortwave

W m^{-2}

LWDOWN

Downward
longwave

W m^{-2}

U2D

10-m eastward wind

m s^{-1}

V2D

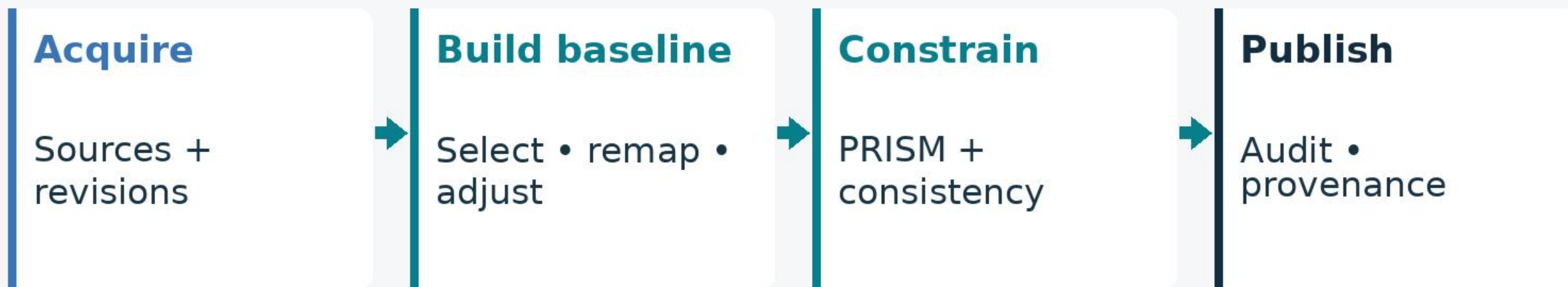
10-m northward
wind

m s^{-1}

Daily file = 24 hourly records. It does not mean daily-resolution data.

One production chain, two published streams

Native source grids map directly to NWM—no common 4-km intermediate grid.



Terrain • masks • cached weights

NRT

Retro

Baseline is shared working data; NRT and Retro remain separate products.

Observation-informed inputs

Nominal source grids and update timing—not delivery guarantees.

Source	Native grid / time	Availability / revisions	Main contribution
MRMS	~1 km / hourly QPE	Pass 1 ~20 min Pass 2 ~60 min*	Storm timing + radar/gauge detail
Stage-IV	~4 km / 1 h + 6 h	Hours; RFC-dependent Later revisions	Regional radar/gauge QC
PRISM	4 km / daily	First ~1 day Stable ~6 months	Rain totals + Tmin/Tmax

*Current MRMS latency. Older versions: ~1 / 2 hours. Actual availability varies.

Complete meteorology across time and space

Complementary roles—not a “reanalysis always wins” rule.

Source	Native grid / time	Availability / cycles	Role in this product
NLDAS-2	0.125° / hourly	~4-day lag	Consistent historical backbone
HRRR	3 km / hourly	Hourly cycles Publication lag varies	Recent hourly analysis; forecast rain
GFS	0.25° / hourly leads	00 / 06 / 12 / 18 UTC Leads 1-6; backup ≤12	Required northern NRT coverage

GFS is a short-forecast gap filler—not an hourly analysis or Retro source.

The same date becomes better informed

Illustrative data age—not a guaranteed product-delivery timeline.



Check actual completeness and revisions—not age alone.

Precipitation: select locally, constrain daily

Quality and coverage can change the preferred source from cell to cell.

Eligible sources

MRMS passes
Stage-IV
NLDAS-2 → HRRR



Hourly composite

Quality + coverage
Regional exceptions
Preserve valid zeros



PRISM constraint

Adjust daily amount
Retain hourly timing
Guard dry/wet cases

GFS fills unsupported northern rain only; it never overwrites valid rain.

PRISM changes the amount—not the storm clock

Schematic example: hourly pattern $\times 1.5$ gives the PRISM daily total.

Hourly composite: 12 mm



PRISM-constrained: 18 mm



PRISM 12-12 windows are internal. Published files remain 00-23 UTC.

Terrain corrections stay physically coupled

Elevation-aware downscaling follows the Cosgrove framework.



PRISM temperature revisions also update humidity and longwave.

Rotate wind vectors correctly. Preserve nighttime shortwave behavior.

The northern HRRR gap must not stop NWM

GFS completion is required for HRRR-selected NRT hours.

GFS: northern coverage

HRRR: native coverage

NLDAS-2 available?

Use NLDAS-2; no GFS needed.

Otherwise: HRRR + GFS

Missing required GFS → preserve old file.

Conceptual footprint—not a geographic map

Keep active NWM cells covered; avoid unnecessary offshore extrapolation.

NRT and Retro answer different questions

Keep both: timeliness can be evaluated against the later stable product.

NRT

What could operations know?

Recent, revisable sources
Early PRISM when eligible
NLDAS replacement as it arrives

Retro

What is our stable reconstruction?

NLDAS-2 backbone
Best eligible precipitation
Stable PRISM constraints

Retro does not overwrite NRT. Record the version used in each simulation.

NRT: refresh → replace only what changed

Dependency fingerprints let unchanged baselines and windows be reused.

Refresh inputs

Check completeness
and source revisions



Resolve changes

NLDAS arrival?
PRISM revision?



Rebuild + audit

Reuse valid work
Replace atomically

Required GFS acquisition is part of the recent-hour production path.

A failed update must not replace accepted data with incomplete output.

Scheduled opportunities $\neq$ current-hour delivery

Repository schedule template • not yet installed on the inspected host

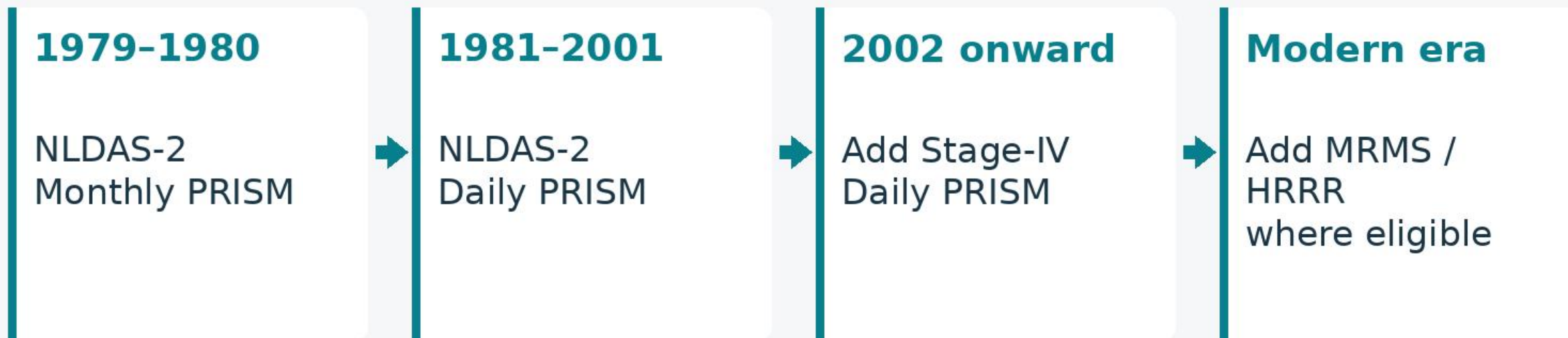
Lane	UTC schedule	Current scan / behavior
NRT	02 / 14 / 20	10 target days; skip unchanged
Daily NRT	08 each day	200-day deeper revision check
Retro	10 on the 18th	45-day eligibility window; stable only

NRT endpoint: UTC today – 2 days. No partial-current-day delivery.

Status report template: every 2 hours • 7-day recent GFS-aware processing tail

Retro: stable constraints, reproducible history

Source availability changes across eras; the common product contract does not.



Stable PRISM → final audit → safe baseline cleanup

Coverage is still being built. Check the inventory—not just the intended era.

One daily file; two clocks to understand

Storage dates and physical accumulation intervals are intentionally separate.

Files on disk

Day D: 00 ... 23 UTC

Day D+1: 00 ...

24-hour NWM run

Reads D 01 ... D+1 00

PRISM rain day

12-12 UTC window → recombine to calendar files

For a full UTC-day simulation, also supply hour 00 from the next file.

Quality safeguards users should know

A complete file is necessary—but not evidence of perfect accuracy.

CNRFC precipitation

Since July 2020: use trusted 6-hour totals, not striped hourly timing.

Coverage envelope

Fill required active cells; control offshore extrapolation.

Physics + publication

Check coupled fields, timestamps, source provenance and file integrity.

Known limits

No universal “best source”; independent validation remains future work.

Fine spacing + stable status + model-read success $\neq$ proven accuracy.

Find, inspect, then use

Consumers need the stream, valid-hour coverage and publication provenance.

**forcing/outputs/conus/{nrt,retro}/YYYY/MM/
YYYYMMDD.LDASIN_DOMAIN1**

Choose stream

NRT for operational
replay
Retro for stable history

Check status

```
python bin/  
report_forcing_status.py
```

Check interval

24 records per day
Next-day 00 for NWM

Save source/version metadata with your experiment. NRT files can change.

What is ready—and what is not yet promised

User expectations as of 19 September 2026

Implemented + tested

- Required GFS northern fallback
- NLDAS arrival / PRISM revisions
- Separate NRT and Retro products
- Failure-safe publication

Still gated

- Live cron coordination
- Consistent sub-hour delivery
- Partial-current-day publication
- Independent accuracy validation

Use the latest inventory for availability; use provenance for reproducibility.

Source guide & further reading

Official provider details and project methods are linked in slide notes.

NOAA / NASA

MRMS • Stage-IV • HRRR • GFS
NLDAS-2 forcing documentation

Oregon State PRISM

Daily update calendar
Grid definitions and revisions

Project documentation

Forcing production workflow
NRT GFS operations + time conventions

Acceptance evidence

NRT revision and reliability reports
Coverage / quality limitations

Provider specifications describe inputs. Check local availability separately.